

ReFuseNet: Refined Feature Fusion for Transferring SAM to Semantic Segmentation

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Abstract

The Segment Anything Model (SAM) provides strong generic visual representations, but adapting it to semantic segmentation remains non-trivial because SAM is not explicitly optimized for class-specific dense prediction. In this work, we present **ReFuseNet**, a refined feature fusion framework for transferring SAM representations to semantic segmentation. Starting from a SAM-based encoder-decoder baseline, we conduct a systematic diagnostic study on how SAM features should be decoded, fused, and refined for dense semantic prediction. Our final model introduces a lightweight CNN image branch that extracts high-resolution detail features from the input image and fuses them with projected final SAM features through an FPN-style pyramid. A lightweight recurrent refinement module is further applied on head-resolution logits using semantic context derived from the SAM final feature. Experiments on CamVid and Cityscapes show that CNN-SAM feature fusion consistently outperforms frozen SAM, low-lr finetuning, pseudo multi-scale decoding, intermediate feature fusion, and a compact Transformer decoder. The refined C4 variant achieves the best performance among our tested SAM-based models, and Cityscapes pretraining followed by CamVid fine-tuning further improves the final result, achieving 84.45 mIoU on CamVid test set. Our analysis suggests that effective SAM transfer requires not only a strong foundation encoder, but also task-specific pipelines that recover local details, preserve semantic guidance, and refine dense predictions.

1. Introduction

Semantic segmentation is a core task in scene understanding, requiring dense pixel-level classification over a fixed set of semantic categories. It is particularly important for autonomous driving and urban perception, where models must distinguish large scene regions such as road, sky, and building, while also preserving thin or small structures

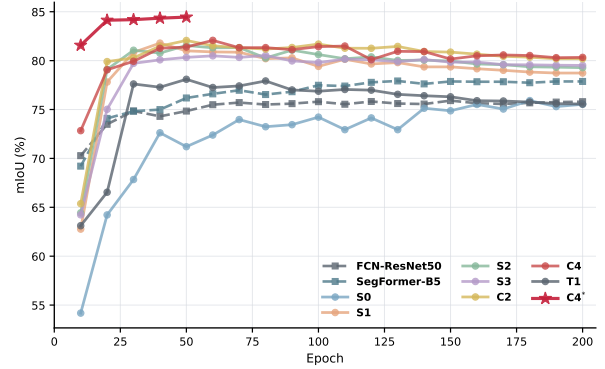


Figure 1. CamVid results for the main variants. ReFuseNet C4 achieves the strongest performance among the tested variants, and C4* further improves after Cityscapes pretraining followed by CamVid fine-tuning.

such as poles, traffic signs, pedestrians, and riders. Recent foundation models have significantly changed the landscape of visual representation learning. The Segment Anything Model (SAM) [5] learns strong generic image representations and supports promptable mask generation across diverse visual domains. However, SAM is not directly designed for fixed-category semantic segmentation. Its masks are promptable and category-independent, while semantic segmentation requires assigning each pixel to a predefined semantic label. Therefore, directly attaching a lightweight classifier to SAM features may not fully exploit SAM’s representation power for dense semantic prediction. In this paper, we study how to adapt SAM to semantic segmentation through a sequence of controlled decoder designs. Rather than proposing a single model in isolation, we build a diagnostic roadmap of SAM-to-segmentation transfer.

Our final model, **ReFuseNet C4**, is motivated by two observations. First, simply resizing the final SAM feature creates a pseudo feature pyramid, but it cannot recover image-level local details that may have been suppressed by the foundation encoder. Second, intermediate SAM features are not necessarily better aligned with fixed semantic la-

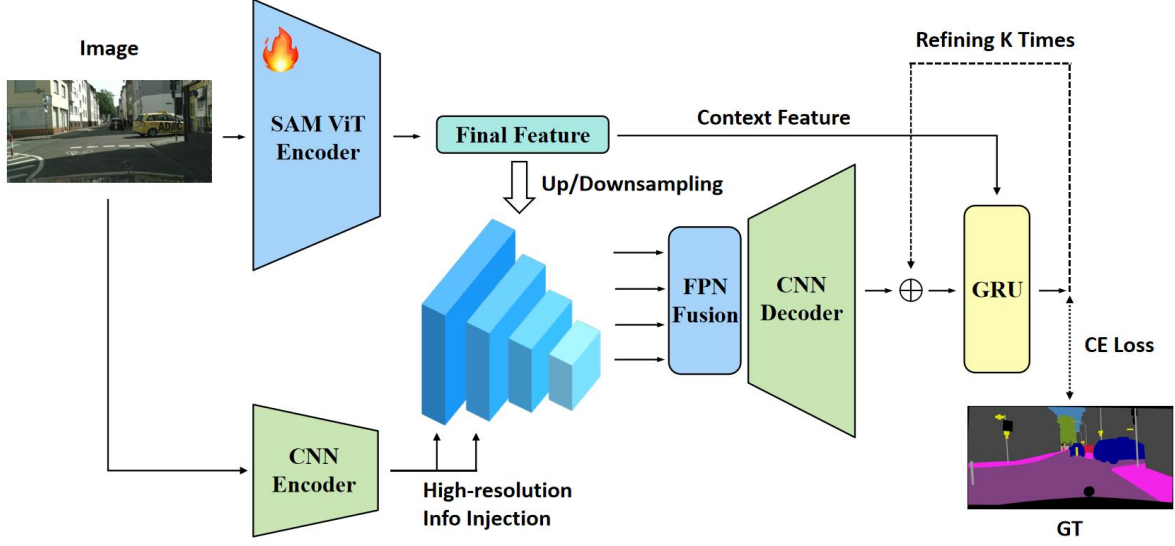


Figure 2. Overview of the ReFuseNet architecture.

bels and may introduce noisy spatial responses. To address these issues, ReFuseNet introduces a lightweight CNN image branch that extracts high-resolution detail features directly from the input image. These image-detail features are fused with projected final SAM features through an FPN-style pyramid, combining local spatial evidence from the image branch with semantic guidance from SAM. Finally, a recurrent refinement module updates head-resolution logits using context derived from the final SAM feature. Our contributions are summarized as follows:

- We provide a diagnostic study of transferring SAM to fixed-category semantic segmentation, covering frozen SAM, low-learning-rate fine-tuning, pseudo multi-scale decoding, intermediate feature fusion, CNN-SAM fusion, Transformer decoding, recurrent refinement, DPT-style decoding, and boundary-aware supervision.
- We develop a CNN-SAM fusion decoder that combines high-resolution image-detail features with projected final SAM features for dense semantic prediction. Built on this fused representation, C4 further improves early convergence through recurrent refinement and achieves the best performance among our tested SAM-based variants.
- We further analyze difficult categories in the appendix by separating rare classes, small-instance classes, and boundary-sensitive classes, showing where boundary supervision and the final ReFuseNet design help or fail.

2. Related Work

Semantic segmentation. Semantic segmentation has been widely studied as dense pixel-level classification. Early fully convolutional networks [8] showed that clas-

sification backbones can be converted into dense predictors by replacing fully connected layers with convolutional prediction heads. Later encoder-decoder and multi-scale designs further improved spatial detail recovery and context aggregation, as seen in architectures such as U-Net [10], DeepLabV3+ [2], and FPN-style feature pyramids [7]. Dense prediction with Vision Transformers has also been explored by DPT [9], which reassembles Transformer tokens into image-like multi-scale representations and progressively fuses them for dense prediction. More recently, Transformer-based segmentation models such as SegFormer [13] and Segmenter [11] have shown strong performance by combining global context modeling with dense prediction heads. In this work, we use FCN-ResNet50 and SegFormer-B5 as external baselines, while our own ablations focus on how SAM features should be adapted for fixed-category segmentation.

Adapting SAM for semantic segmentation. SAM is designed for promptable and class-agnostic mask prediction rather than fixed-category semantic segmentation. Several works have explored how to introduce semantic information into SAM-style segmentation. Semantic-SAM [6] extends SAM toward semantic-aware and multi-granularity segmentation; SAM-CLIP [12] combines SAM’s spatial segmentation ability with CLIP’s semantic recognition for zero-shot semantic segmentation; and SAM-Path [14] adapts SAM to pathology semantic segmentation using trainable class prompts and a domain-specific foundation encoder. Other works use SAM to improve semantic pseudo-labels in weakly supervised semantic segmentation,

exploiting SAM’s object-aware masks to refine class-aware CAM regions [3]. Rather than aiming to build a general-purpose semantic SAM model or an open-vocabulary segmentation system, this paper studies a narrower and more controlled setting: adapting SAM features to fixed-category urban-scene semantic segmentation.

3. Methods

3.1. Problem Setting

Given an input image $I \in \mathbb{R}^{3 \times H \times W}$, the goal of semantic segmentation is to predict a dense label map $\hat{Y} \in \{1, \dots, K\}^{H \times W}$ over K predefined classes. We use SAM ViT-B as the visual encoder and design different decoders to convert SAM features into class-specific semantic logits. All models are trained with raw logits using cross-entropy loss:

$$\mathcal{L}_{seg} = \text{CE}(Z, Y), \quad (1)$$

where $Z \in \mathbb{R}^{K \times H \times W}$ denotes the predicted logits and pixels with label 255 are ignored.

3.2. SAM-based Segmentation Baselines

We first construct several SAM-based diagnostic baselines.

Frozen-SAM (S0). S0 freezes the SAM encoder and attaches a lightweight decoder to the final SAM feature. This baseline tests whether the generic SAM representation is directly sufficient for fixed-category semantic segmentation.

FT-SAM (S1). S1 fine-tunes the SAM encoder using a low learning rate while keeping the same final-feature decoder. This setting evaluates whether SAM features require task-specific adaptation for semantic labels.

Pseudo-Pyramid (S2). S2 constructs a pseudo multi-scale feature pyramid by resizing the final SAM feature to multiple resolutions. These resized features are fused by a lightweight decoder. S2 tests whether multi-scale decoding is useful even when all scales originate from the same final SAM representation.

Intermediate-Fusion (S3). S3 uses intermediate SAM features for multi-scale fusion. This variant examines whether earlier SAM features provide useful dense information for semantic segmentation. Empirically, S3 is weaker than S2 in our experiments, suggesting that intermediate SAM features may introduce spatial noise or features not aligned with fixed-category segmentation.

3.3. CNN-SAM Feature Fusion

The main limitation of S2 is that its multi-scale features are generated by resizing the same final SAM feature map. This creates a pseudo pyramid, but all pyramid levels still originate from the same semantic representation. As a result, S2 lacks an explicit mechanism for recovering high-resolution image details. To address this, we introduce a lightweight CNN image branch in C2 and C4. Given the preprocessed input image I and the final SAM feature F_{sam} , the CNN branch extracts two high-resolution detail features:

$$\{F_{img}^{256}, F_{img}^{128}\} = \text{CNN}_{img}(I). \quad (2)$$

In parallel, the final SAM feature is projected and resized to multiple pyramid levels:

$$F_{sam}^l = \text{Resize}_l(\text{Proj}_l(F_{sam})). \quad (3)$$

At high-resolution levels, we concatenate image-detail features with the resized SAM features:

$$P^l = \phi_l(\text{Concat}(F_{img}^l, F_{sam}^l)), \quad l \in \{256, 128\}. \quad (4)$$

At lower-resolution levels, projected SAM features provide semantic guidance directly. The resulting pyramid is fused in a top-down FPN-style manner to produce the head-resolution semantic feature:

$$F_{sem} = \text{FPN}(P^{32}, P^{64}, P^{128}, P^{256}). \quad (5)$$

We denote this CNN-SAM fusion variant as **C2**. Compared with S2, C2 does not rely only on resized SAM features. Instead, it explicitly injects image-level local detail through the CNN branch while preserving high-level semantic guidance from SAM. This design explains why C2 provides the largest improvement in the Cityscapes roadmap.

3.4. Logit Refinement

As illustrated in Figure 2, our final model, **ReFuseNet C4**, extends C2 with a lightweight recurrent refinement module after the CNN-SAM fusion decoder. The refinement operates on head-resolution logits rather than full-resolution logits for efficiency. Let Z_t be the intermediate logits at refinement step t . The refinement module updates the logits using semantic context derived from the final SAM representation:

$$Z_{t+1} = \text{GRU}(Z_t, C_{sam}), \quad (6)$$

where C_{sam} is obtained from a projection of the final SAM feature. This design avoids stacking intermediate SAM features as refinement context and keeps the refinement guided by the final high-level SAM representation. The final logits are then upsampled to the target resolution and supervised by cross-entropy loss. In our experiments, C4 provides a small but consistent improvement over C2 and also shows stronger early-stage evaluation performance.

Table 1. Main results on CamVid and Cityscapes. C4 denotes the final ReFuseNet model. C4* denotes Cityscapes pretraining followed by CamVid fine-tuning.

Method	Params (M)	CamVid		Cityscapes	
		mIoU (%)	Pixel Acc. (%)	mIoU (%)	Pixel Acc. (%)
FCN-ResNet50	32.95M	75.91	94.81	73.50	95.55
SegFormer-B5	84.60M	77.92	95.60	80.74	<u>96.51</u>
ReFuseNet S0	90.15M	75.91	95.04	59.11	93.95
ReFuseNet S1	90.16M	81.79	96.39	78.14	96.35
ReFuseNet S2	90.25M	81.55	<u>96.42</u>	78.27	<u>96.46</u>
ReFuseNet S3	90.52M	80.53	96.26	77.42	96.36
ReFuseNet C2	91.80M	<u>82.06</u>	<u>96.58</u>	<u>78.69</u>	96.58
ReFuseNet C4	92.07M	<u>82.07</u>	96.40	<u>78.72</u>	96.58
ReFuseNet T1	93.03M	78.09	95.90	76.43	96.04
ReFuseNet C4*	92.07M	84.45	96.87	–	–

3.5. Transformer Decoder Variant

We additionally implement a compact Segmenter-style Transformer decoder, denoted as **T1**. T1 replaces the CNN/FPN decoder with a mask Transformer operating on final SAM tokens. The final SAM feature is projected into patch tokens, concatenated with learnable class tokens, and processed by several Transformer blocks. The mask logits are produced by patch-class similarity:

$$M = \hat{P}\hat{C}^\top, \quad (7)$$

where $\hat{P}$ and $\hat{C}$ are normalized patch and class token embeddings. T1 is included as a decoder ablation to test whether compact global token interaction can replace CNN-based dense detail reconstruction. Empirically, T1 underperforms C2 and C4, suggesting that recovering local spatial details is critical for adapting SAM to semantic segmentation.

3.6. Supplementary Analysis

In addition to the main results, the appendix provides three diagnostic analyses. First, we report S4, S5, and S6, where S4 adds GRU refinement to the S2 pseudo-pyramid decoder, S5 uses a DPT-style decoder head, and S6 adds an auxiliary boundary head to S1. These variants help separate the effects of refinement, decoder strength, and boundary supervision. Second, we visualize SAM spatial attention before and after task-specific training to show that only the final SAM layer approaches the image-wide attention pattern that per-pixel prediction requires, while intermediate layers remain scattered and salient-region-focused — which directly explains why naive intermediate fusion (S3) hurts. Third, we analyze rare and small-instance categories using dataset-level pixel frequency and connected-component statistics. This analysis clarifies whether improvements come mainly from frequent large-region classes or also benefit difficult rare and small categories.

4. Experiments

4.1. Datasets and Evaluation

We evaluate on CamVid [1] and Cityscapes [4]. CamVid contains urban driving scenes with 11 semantic classes, while Cityscapes contains high-resolution street-scene images with 19 trainId classes. Void pixels are ignored with label 255. For CamVid, models are evaluated on whole images. For Cityscapes, we use sliding-window evaluation with crop size 1024 and stride 768, following prior works [13]. We evaluate using mean Intersection-over-Union (mIoU).

4.2. Main Comparison

Table 1 reports the main comparison. We compare two external baselines, FCN-ResNet50 and SegFormer-B5, with our SAM-based variants S0, S1, S2, S3, C2, C4, and T1. The parameter column makes the comparison more transparent: improvements are not solely a consequence of larger models, since several stronger variants remain in the same parameter range as their predecessors.

The results show several trends. First, S0 performs poorly on Cityscapes compared with fine-tuned variants, reaching only 59.11 mIoU versus 78.14 for S1. On CamVid, the same jump is 75.91 to 81.79. This gap shows that directly using frozen SAM features is insufficient for fixed-category semantic segmentation. Second, S1 improves over S0 by +5.88 mIoU on CamVid and +19.03 mIoU on Cityscapes, which confirms that low-learning-rate SAM adaptation is essential. Third, S2 slightly improves the Cityscapes result over S1 by +0.13 mIoU, while S3 falls back by -0.85. Fourth, C2 improves over S2 by +1.27 mIoU on Cityscapes and +0.51 on CamVid, which is the main jump in the roadmap. Finally, C4 achieves the best result among the SAM-based variants on both datasets, with 82.07 mIoU on CamVid and 78.72 mIoU on Cityscapes, showing that logit refinement provides only a small gain on top of a stronger fused representation.

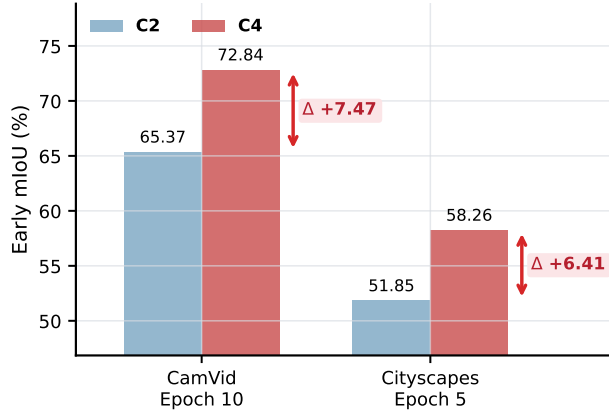


Figure 3. First-evaluation comparison between C2 and C4.

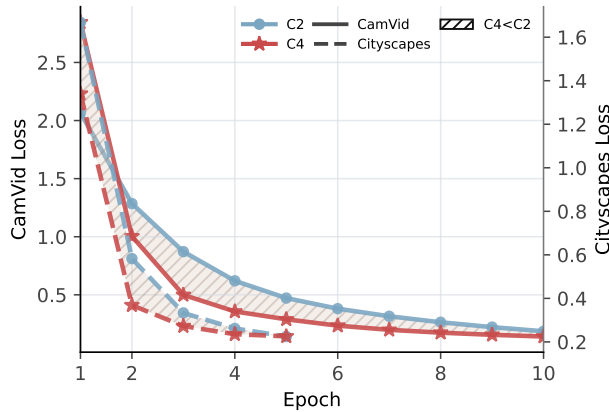


Figure 4. Early training loss comparison between C2 and C4.

Although C4 only slightly improves the final best mIoU over C2, it shows clearer benefits during early training. Figure 4 compares the early training loss of C2 and C4. C4 decreases the training loss faster on both CamVid and Cityscapes, suggesting that the recurrent refinement module helps stabilize optimization in the early stage. This trend is also reflected in the first evaluation results. As shown in Figure 3, C4 improves the first evaluation mIoU from 65.37 to 72.84 on CamVid (+7.47), and from 51.85 to 58.26 on Cityscapes (+6.41). However, the final best results of C2 and C4 remain close, differing by only +0.01 on CamVid and +0.03 on Cityscapes. This indicates that the refinement module mainly improves early prediction stability and convergence behavior, rather than substantially raising the final performance ceiling.

4.3. Diagnostic Roadmap on Cityscapes

To better understand the contribution of each design step, we summarize the Cityscapes roadmap from S1 to C4 in Table 2. Starting from S1, pseudo multi-scale decoding pro-

Table 2. Cityscapes diagnostic roadmap from S1 to C4.

Step	Design	Cityscapes mIoU (%)
S1	SAM low-LR fine-tune	78.14
S2	Pseudo multi-scale	78.27 (+0.13)
S3	Intermediate features	77.42 (-0.85)
C2	CNN detail fusion	78.69 (+1.27)
C4	CNN fusion + refine	78.72 (+0.03)

vides a small +0.13 mIoU improvement. Intermediate feature fusion decreases performance by -0.85, indicating that earlier SAM features may not be well aligned with semantic labels. CNN detail fusion provides the largest gain among the tested modifications at +1.27, and refinement further brings a small +0.03 gain. The roadmap is useful because it separates representational improvement from optimization-side improvement: C2 changes the feature space, while C4 mainly improves how that space is used during prediction.

4.4. Cityscapes Pretraining and CamVid Fine-tuning

We further evaluate whether the best ReFuseNet model can benefit from large-dataset pretraining. We pretrain models on Cityscapes and then fine-tune them on CamVid. As shown in Figure 5, ReFuseNet C4 improves from 82.07 to 84.45 mIoU after Cityscapes pretraining followed by CamVid fine-tuning, for a gain of +2.38. This suggests that the proposed decoder benefits from source-domain urban-scene supervision before adapting to the smaller CamVid dataset. The effect is modest in absolute terms, but it supports the practical value of using Cityscapes as a source dataset before fine-tuning on CamVid.

4.5. Complementary Analysis

Failure Modes. Taken together, the weaker variants point to two recurring failure modes. S3 fails because SAM’s intermediate attention is not the kind of attention that per-pixel semantic segmentation needs: even in the raw model the final layer is already the most uniformly distributed across the image, and S2 training tightens it further toward the image-wide distribution suitable for dense prediction, while the intermediate layers stay scattered and salient-region-focused throughout. Fusing intermediate features back into the decoder therefore injects a non-uniform attention signal into a path that already has the right one, and the extra intermediate-fusion branches also disrupt the optimization that drives the final SAM layer toward uniform attention. Appendix Sec. C visualizes this directly and shows that the CNN image branch in C2 and C4 addresses a separate problem — high-resolution detail recovery — rather than fixing SAM’s attention distribution. T1 shows the opposite issue: a compact Transformer decoder

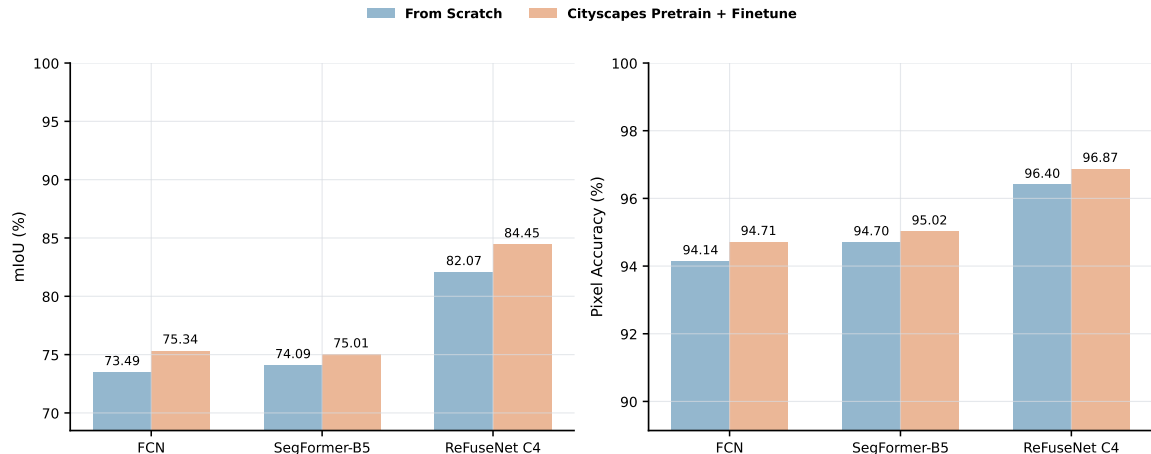


Figure 5. From-scratch training versus Cityscapes pretraining followed by CamVid fine-tuning.

can mix tokens globally, but without an explicit image-style multi-scale reconstruction path it still struggles to recover thin structures and small objects. The appendix analysis of the DPT-style ablation in Appendix Sec. B.2 shows a related pattern: reorganizing SAM tokens into a pyramid is not the same as recovering native image-derived local detail.

Overfitting. The CamVid curve suggests that C4 reaches its best point earlier than the baselines. The stronger decoder seems to adapt faster to the small training set, while the baselines remain flatter but lower in final accuracy.

Cityscapes Performance. SegFormer-B5 still leads on Cityscapes, which is reasonable because it is a segmentation-native model with a stronger multi-scale decoder, while our roadmap is still centered on SAM transfer and local-detail reconstruction.

5. Conclusion

We presented ReFuseNet, a refined feature fusion framework for adapting SAM to fixed-category semantic segmentation. Across a controlled roadmap of variants, we found that frozen SAM features are not enough, low-learning-rate adaptation is important, and naive intermediate fusion can degrade performance. The strongest gains come from CNN-SAM feature fusion, where an image branch reconstructs local detail before semantic prediction, while the recurrent refinement module mainly improves early convergence and prediction stability. Cityscapes pretraining further improves the final model on CamVid, but the gain over C2 remains modest. Overall, the results suggest that successful SAM transfer depends more on explicit local reconstruction than on simply rearranging the original SAM representation.

Limitations. We only evaluate SAM ViT-B on urban-scene datasets, so broader generalization is still open. The boundary head is a weak auxiliary signal rather than a full fusion path, and the current method does not explicitly optimize long-tail imbalance. We also keep the Transformer decoder compact, so it is a diagnostic baseline rather than a full-strength alternative.

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Appendix

A. Implementation Details and Variant Definitions

A.1. Model Variants

Table 3 summarizes the model variants evaluated in this project. All SAM-based variants use SAM ViT-B as the visual encoder. The main paper focuses on S0–S3, C2, C4, and T1. In the appendix, we additionally report S4, S5, and S6 as diagnostic variants.

Table 3. Definition of ReFuseNet variants. S4–S6 are appendix-only diagnostic variants.

Variant	Params (M)	Description
S0	90.15	Frozen SAM encoder + final SAM feature decoder
S1	90.16	Low-learning-rate SAM fine-tuning + final feature decoder
S2	90.25	S1 + pseudo multi-scale pyramid from the final SAM feature
S3	90.52	S1 + intermediate SAM feature fusion
S4	90.52	S2 + GRU refinement on head-resolution logits
S5	100.25	S1 + DPT-style decoder head
S6	90.30	S1 + auxiliary boundary head
C2	91.80	Final SAM feature + lightweight CNN image branch + FPN fusion
C4	92.07	C2 + GRU refinement with SAM-final-only context
T1	93.03	Final SAM feature + compact Segmenter-style Transformer decoder

A.2. Training Hyperparameters

Table 4 lists the core hyperparameters used in the main experiments. We keep the optimization recipe intentionally simple so that differences between variants are attributable to decoder design rather than training tricks.

Across both datasets, we use gradient accumulation with four steps, polynomial learning-rate decay, and a boundary auxiliary loss weight of 0.1 for S6. Cityscapes uses a longer warmup because the higher-resolution crops make the optimization noisier during the first few iterations. Here, global steps denote the total number of optimizer updates after gradient accumulation.

A.3. Training and Evaluation

All models are trained with raw logits using cross-entropy loss:

$$\mathcal{L}_{seg} = \text{CE}(z, y), \quad (8)$$

where pixels with label 255 are ignored. CamVid models are trained for 200 epochs, while Cityscapes models are trained for 50 epochs. For Cityscapes, we use sliding-window evaluation with crop size 1024 and stride 768.

We evaluate using mean Intersection-over-Union (mIoU). A full validation-set confusion matrix $M \in \mathbb{R}^{K \times K}$ is accumulated, where rows represent ground-truth classes and columns represent predicted classes. For class c ,

$$\text{IoU}_c = \frac{M_{cc}}{\sum_j M_{cj} + \sum_i M_{ic} - M_{cc}}. \quad (9)$$

Classes with zero union are treated as NaN, and mIoU is computed over valid classes.

A.4. Cityscapes Training Curve

The Cityscapes training curves are moved here to keep the main paper focused on the final comparison and the refinement analysis. The curve is useful mainly as a diagnostic sanity check: it shows that the ranking between variants stabilizes early, while the later epochs mostly refine small margins among the stronger models.

Table 4. Training hyperparameters used in the main experiments. The same optimizer family and loss recipe are shared across variants.

Setting	Value
CamVid	
Epochs	200
Train bs	2
Eval bs	4
Grad accum	4
Optimizer	AdamW
Global steps	9200
LR (dec/SAM)	1e-4 / 1e-5
WD	1e-4
Crop / inference	360×480 / whole
Cityscapes	
Epochs	50
Train bs	2
Eval bs	1
Grad accum	4
Optimizer	AdamW
Global steps	18600
LR (dec/SAM)	1e-4 / 1e-5
WD	1e-4
Crop / inference	1024×1024 / sliding 1024×1024, stride 768

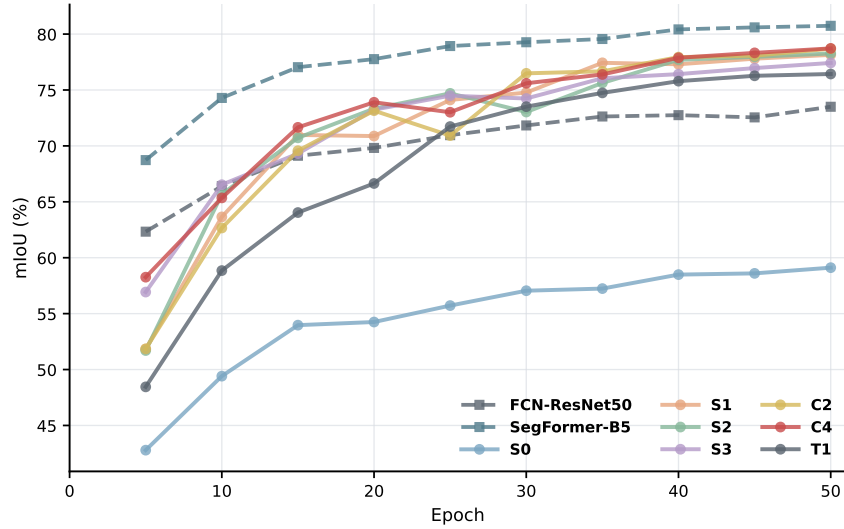


Figure 6. Cityscapes mIoU curves of the baselines and ReFuseNet variants.

B. Additional Decoder Ablations

B.1. S4: Recurrent Refinement on the Pseudo Pyramid

S4 adds GRU refinement on top of S2. Different from C4, which refines logits after CNN-SAM feature fusion, S4 directly refines the S2 pseudo-pyramid prediction. This ablation tests whether recurrent refinement alone is sufficient to improve the early-stage segmentation behavior.

As shown in Fig. 7, S4 reduces the early training loss compared with S2 on both CamVid and Cityscapes. This suggests that recurrent refinement helps optimization and accelerates early convergence.

However, faster convergence does not necessarily imply a better final model. Fig. 7 shows that S4 substantially improves

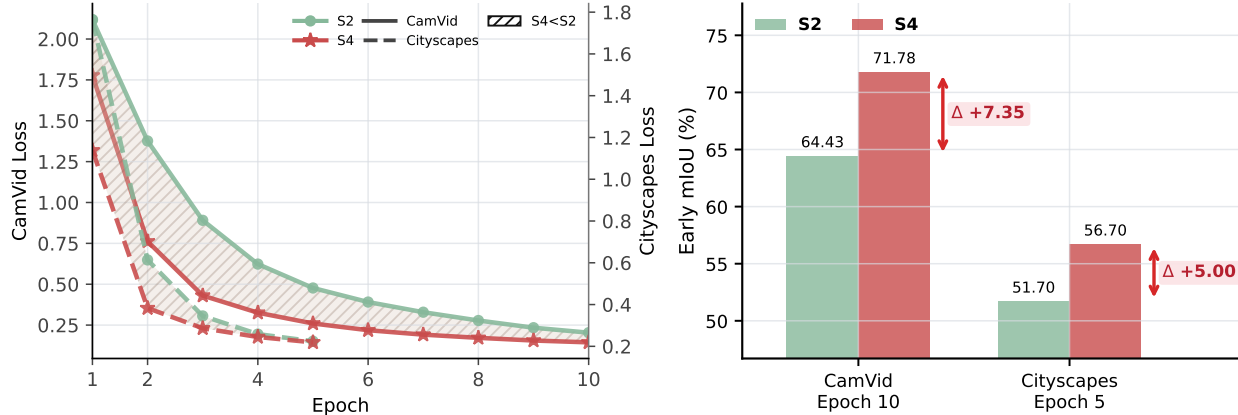


Figure 7. Early training loss comparison between S2 and S4. S4 adds GRU refinement to S2 and converges faster in the early training stage.

the first evaluation result on both datasets: CamVid goes from 64.43 to 71.78 mIoU, and Cityscapes goes from 51.70 to 56.70 mIoU. This means refinement mainly changes the early optimization trajectory rather than just the final checkpoint.

This result motivates the design of C4 in the main paper. Instead of applying refinement directly on the pseudo-pyramid decoder, C4 first improves the feature representation through CNN-SAM fusion and then applies lightweight recurrent refinement. In other words, refinement is more effective when the underlying semantic feature is stronger.

B.2. S5 and S6: DPT Head and Boundary Head

We further evaluate two appendix-only decoder variants. S5 replaces the simple final-feature decoder with a DPT-style decoder head. S6 adds an auxiliary boundary head to S1. The goal of S5 is to test whether a stronger decoder head improves SAM transfer, while the goal of S6 is to test whether boundary supervision can help dense semantic prediction.

Table 5. Appendix-only ablation results for S5 and S6. S5 uses a DPT-style head, while S6 adds an auxiliary boundary head to S1.

Dataset	Method	Params (M)	Epoch	mIoU (%)	Pixel Acc. (%)
CamVid	S4	90.52M	40	81.09	96.17
CamVid	S5	100.25M	110	80.19	96.21
CamVid	S6	90.30M	80	80.70	96.30
Cityscapes	S4	90.53M	45	78.12	96.44
Cityscapes	S5	100.25M	50	77.87	96.52
Cityscapes	S6	90.30M	50	77.21	96.44

The results show that S5 and S6 are competitive but do not outperform the final C2/C4 variants reported in the main paper. This suggests that simply replacing the decoder head or adding auxiliary boundary supervision is not sufficient. The strongest gains come from explicitly reconstructing dense CNN detail features and fusing them with projected SAM semantic features.

For S5 specifically, the DPT-style head is stronger than a plain final-feature decoder in the sense that it reassembles Transformer features into a top-down pyramid. However, it still works on features that all originate from the same SAM encoder output, so the pyramid is an internal re-organization of semantic tokens rather than a genuinely new image-derived multi-scale representation. In other words, DPT improves decoder structure, but it does not inject the kind of high-resolution local detail that a native CNN image branch can provide. This helps explain why S5 is reasonable as a diagnostic baseline but still falls short of C2 and C4, where local detail is reconstructed explicitly before fusion.

For S6, the boundary head is treated as a diagnostic pilot rather than a main positive result. A boundary branch can encourage high-frequency spatial awareness, but if the boundary prediction is only supervised as an auxiliary output and not fused back into the semantic logits, its effect on final segmentation is indirect. In addition, boundary pixels are highly imbalanced, so naive boundary supervision can be dominated by large-region class transitions rather than rare small structures.

C. SAM Feature Visualization

To understand why intermediate SAM feature fusion does not help (S3 in Table 1), we inspect *where* SAM actually attends, both before and after task-specific training. Figure 8 plots the spatial attention response of raw SAM ViT-B at blocks 3, 6, 9, 12, with no segmentation training applied. Figure 9 plots the same probe after training S2 on CamVid/Cityscapes. For each layer we average attention weights over heads and query tokens and overlay the result on the input image; a bright region indicates that many spatial tokens route attention through that location.

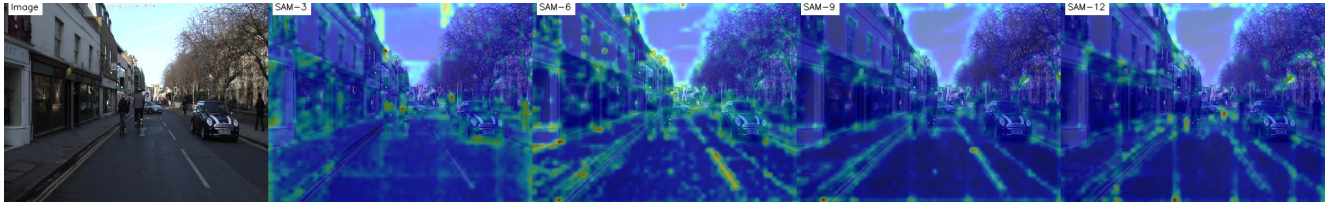


Figure 8. Raw SAM ViT-B attention at blocks 3, 6, 9, 12, before any segmentation training. The earlier layers (3, 6, 9) are scattered and salient-region-focused: they concentrate on sky boundaries, building edges, road markings, and high-contrast vegetation while leaving large parts of the scene dim. The final layer (12) is already noticeably more uniformly distributed across the image — leaning toward, but not yet reaching, the image-wide attention pattern that per-pixel semantic segmentation requires.



Figure 9. Same attention probe after training S2 on CamVid/Cityscapes. The earlier layers (3, 6, 9) still exhibit the same scattered, salient-region-focused pattern as in the raw model and do not become image-wide during training. The final layer (12) becomes even more uniformly distributed across the whole image — the image-wide response required by per-pixel prediction is now essentially achieved. The two rows together show that training pushes only the final SAM layer toward the right attention distribution; the intermediate layers remain non-uniform throughout.

The two rows together explain the S3 failure. Semantic segmentation assigns a label to every pixel, so its ideal spatial attention is roughly image-wide. The final SAM layer already leans toward this uniform, image-wide distribution in the raw model, and S2 training tightens it further; only it carries the right kind of response for dense prediction. The intermediate SAM features, by contrast, remain scattered and salient-region-focused at every training stage. Fusing them into the decoder (S3) therefore injects a non-uniform attention signal into a path that already has a uniform one, and the additional intermediate-fusion branches also disrupt the optimization that drives the final SAM layer toward the image-wide distribution. S3 fails not because intermediate features are uninformative, but because their attention distribution is the wrong shape for the task, and forcing the network to align them with the final uniform feature interferes with the training process.

Why this is separate from the CNN image branch. The CNN image branch in C2 and C4 addresses a different problem from the one visualized above. Even when SAM’s final feature attends image-wide, it still operates at low spatial resolution and cannot directly recover thin structures or small instances. The CNN branch therefore does not try to “fix” SAM’s attention distribution — it injects an explicit high-resolution signal directly from the input image so that the decoder has the spatial detail it needs on top of SAM’s semantic guidance.

D. Rare and Small-instance Categories

Semantic segmentation performance is not uniform across classes. Frequent large-region classes such as road, building, sky, and vegetation are usually easier, while low-frequency and small-instance classes remain challenging. All long-tail numbers below are computed from the `last.pth` checkpoints.

Table 6. Grouped S1 performance on rare and small-instance categories. Higher values indicate better class-level IoU.

Dataset	Group	#Cls	S1	S6	C4	S6-S1	C4-S1
CamVid	All	11	78.73	78.62	80.33	-0.11	+1.60
CamVid	Rare	4	59.86	59.81	63.67	-0.05	+3.81
CamVid	Non-rare	7	89.51	89.36	89.86	-0.15	+0.34
CamVid	Small	4	56.45	56.44	61.01	-0.01	+4.56
CamVid	Non-small	7	91.46	91.29	91.38	-0.18	-0.08
CamVid	Rare $\cap$ Small	3	52.30	52.53	57.79	+0.23	+5.49
Cityscapes	All	19	78.14	77.21	78.72	-0.93	+0.58
Cityscapes	Rare	10	70.75	68.73	71.07	-2.02	+0.32
Cityscapes	Non-rare	9	86.34	86.62	87.21	+0.28	+0.87
Cityscapes	Small	6	71.56	71.86	73.66	+0.30	+2.10
Cityscapes	Non-small	13	81.17	79.67	81.05	-1.50	-0.12
Cityscapes	Rare $\cap$ Small	4	70.36	70.46	72.35	+0.09	+1.99

Table 7. Class statistics used to define rare and small-instance categories. Daggers are appended directly to the class name: † rare, ‡ small-instance, and †‡ both. All long-tail statistics are computed from `last.pth` checkpoints.

Dataset	Class	Pixel freq. (%)	Image freq. (%)	Mean comp./inst. size (%)
CamVid	Bicyclist†‡	0.30	54.77	0.40
CamVid	Pedestrian†‡	0.67	86.38	0.17
CamVid	Pole†‡	1.02	99.73	0.05
CamVid	Fence†	1.17	47.14	0.52
CamVid	SignSymbol‡	1.22	95.64	0.15
CamVid	Sky	17.54	99.73	1.60
CamVid	Building	24.21	99.46	2.66
CamVid	Road	32.96	100.00	9.96
CamVid	Pavement	4.67	95.10	0.84
CamVid	Tree	10.12	86.92	0.59
CamVid	Car	6.11	98.09	1.30
Cityscapes	rider†‡	0.13	34.39	0.17
Cityscapes	traffic light†‡	0.21	55.73	0.06
Cityscapes	bicycle†‡	0.41	55.33	0.19
Cityscapes	traffic sign†‡	0.55	94.39	0.10
Cityscapes	motorcycle†	0.10	17.24	0.29
Cityscapes	train†	0.23	4.77	1.51
Cityscapes	bus†	0.24	9.21	1.02
Cityscapes	truck†	0.27	12.07	1.14
Cityscapes	wall†	0.66	32.61	0.38
Cityscapes	fence†	0.88	43.56	0.42
Cityscapes	pole‡	1.23	99.13	0.07
Cityscapes	person‡	1.22	78.76	0.27
Cityscapes	road	36.88	98.62	5.74
Cityscapes	sidewalk	6.09	94.49	0.68
Cityscapes	building	22.82	98.62	1.21
Cityscapes	vegetation	15.92	97.18	1.74
Cityscapes	terrain	1.16	55.60	0.36
Cityscapes	sky	4.01	90.29	1.23
Cityscapes	car	7.00	95.19	1.25

We distinguish two concepts. Rare classes are defined by low training-set pixel frequency. Small-instance classes are identified by small average connected-component or instance size. These two groups are related but not identical, which is why we mark the class names directly with † for rare classes, ‡ for small-instance classes, and †‡ for classes that are both.

The grouped numbers in Table 6 and the two class-level plots together show a consistent pattern. Rare classes are substantially harder than non-rare classes on both datasets, but the effect is sharper on CamVid, where S1 reaches 59.86 mIoU on rare classes versus 89.51 on non-rare classes. Cityscapes shows the same gap, with S1 at 70.75 on rare classes versus 86.34 on non-rare classes. The plots in Fig. 10 make the same point visually: the low-frequency classes sit well below the dominant classes, even when the model is strong on the overall benchmark.

The small-instance analysis is similar but not identical. Table 7 shows that the classes selected as small are not just rare by pixel count; they also have small connected components or instances, which makes them sensitive to boundary precision and local detail recovery. In Table 6, S6 does not materially improve small classes over S1, especially on CamVid where the change is 56.45 to 56.44, while C4 raises the small-class average to 61.01. On Cityscapes the same trend appears more gently: S1 is 71.56, S6 is 71.86, and C4 is 73.66. Fig. 11 shows that these gains are concentrated in thin or fragmented categories rather than in large background regions.

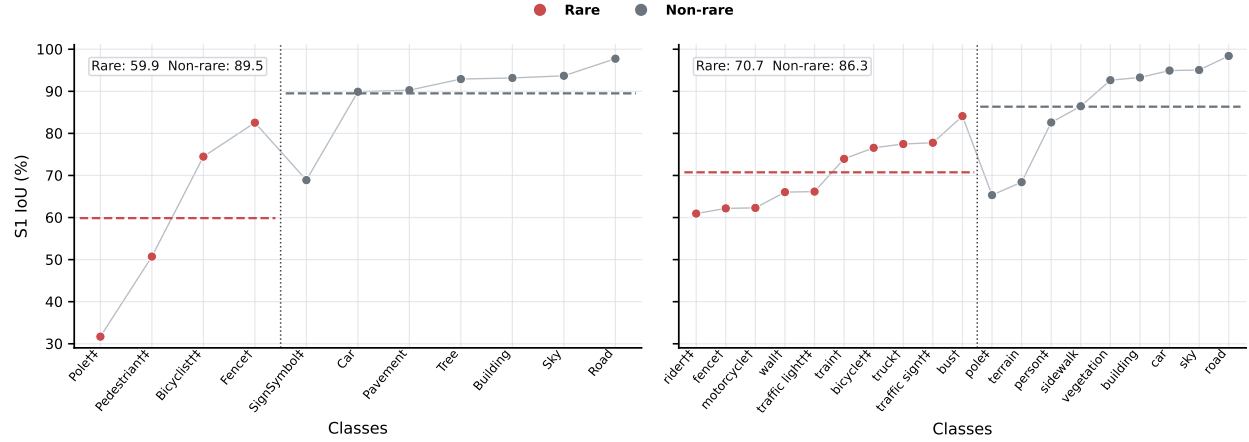


Figure 10. S1 per-class IoU grouped by rare and non-rare classes. The rare classes are consistently harder in both CamVid and Cityscapes.

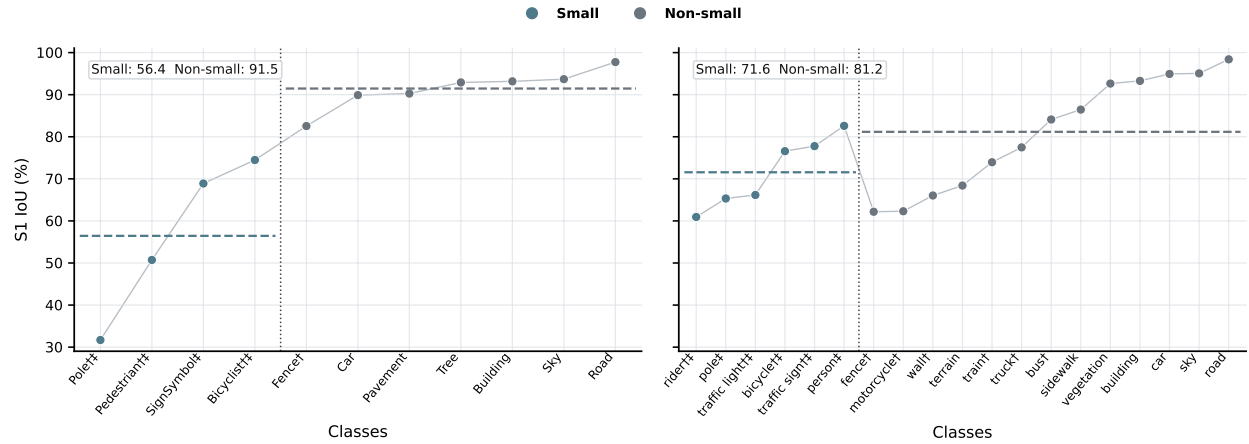


Figure 11. S1 per-class IoU grouped by small-instance and non-small classes. Small-instance categories are generally less accurate, especially for thin or fragmented structures.

Taken together, the appendix long-tail analysis supports the main design claim of ReFuseNet. Auxiliary boundary supervision alone is not enough to resolve long-tail difficulty, because the semantic head still lacks stronger local reconstruction. The CNN-SAM fusion path in C4 is more effective because it injects a richer spatial decoder before prediction, which helps both rare classes and small-instance categories.

E. Qualitative Example

Fig. 12 shows a CamVid scene predicted by ReFuseNet C4. The prediction preserves coherent large semantic regions (road, building, sky, vegetation) while recovering several small or thin street-scene structures.

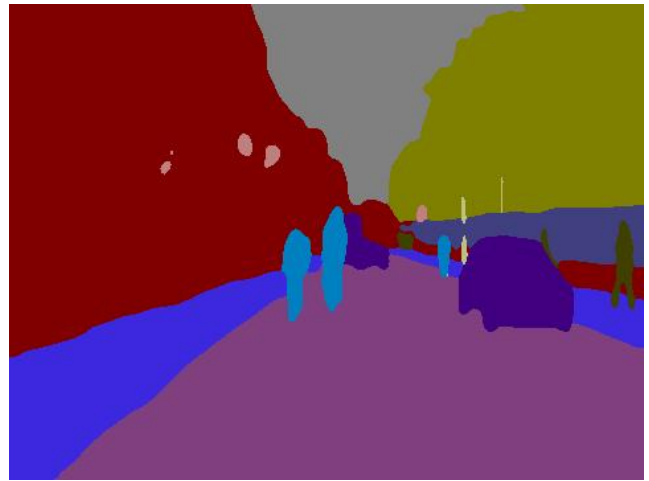


Figure 12. Qualitative example on CamVid. Left: input image. Right: ReFuseNet prediction.